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$\sigma_i(A)$ Formula

First note that the determinant is multi-linear, so we can write

$$A = \begin{pmatrix} a_{11} - x & \cdots \\ \cdots & a_{nn} - x \end{pmatrix} = \begin{pmatrix} a_{11} & \cdots \\ \cdots & a_{nn} \end{pmatrix} - \begin{pmatrix} x & \cdots \\ \cdots & a_{nn} \end{pmatrix} - \begin{pmatrix} x & \cdots \\ \cdots & a_{nn} \end{pmatrix} + \begin{pmatrix} x & \cdots \\ \cdots & x \end{pmatrix}$$

Assume we have a single $a_{ii} = x$. We are only interested in the coefficient for x. Fixing π and using the Leibniz formula, we get

$$k = -x \sum_{\pi \in S(n) : \pi(i) = i} \operatorname{sgn}(\pi) a_{1\pi(1)} \cdots a_{(i-1)\pi(i-1)} a_{(i+1)\pi(i+1)} \cdots a_{n\pi(n)}$$

Using the same argument as for the Laplace expansion, when we have P_π, and delete the ith column and row, then $P'_\pi = P_{\pi'}$ corresponds to a valid $\pi' \in S(n-1)$. Then $\operatorname{sgn}(\pi) = \operatorname{sgn}(\pi')$, since the element at $\pi(i) = i$ is already in the identity position.

$$k = -x \det(A_{ii})$$

Using the multilinearity we see

$$\sigma_1(A) = -x \sum_{i=1}^{n} \det(A_{ii})$$

We can generalize the argument above, by fixing $\pi(i_1) = i_1, ..., \pi(i_m) = i_m$. Then we sum over the permutations for the remaining minor matrices $A_{(n-m)(n-m)}$. Note that in the multilinear expansion, the leading unit is $(-1)^m$.

Thus we get the generalized formula for characteristic polynomial.

Exercises 8.1

Ex 8.1.10: Matrices with Same Charpoly are Similar

Let $B = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$, then charpoly$(B)$ = charpoly(I) but they are not similar since $S = PIP^{-1} = I$.

Ex 8.1.12: 0 is an Eigenvalue of $A \iff A$ is Singular

This means the constant term of charpoly is 0. Therefore there is a row i with $(A - xI)_{ii} = -x$ and all other entries 0. Which means A has a row of all zeroes, and hence is singular.

Ex 8.1.15: A and $-A$ are Similar

Prove $\det(A) = \operatorname{Tr}(A) = 0$. Note $A = -PAP^{-1} \implies \det(A) = -\det(P)\det(A)\det(P^{-1}) = -\det(A) \implies \det(A) = 0$. Then using $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$, we can do $\operatorname{Tr}(A) = \operatorname{Tr}(-PAP^{-1}) = -\operatorname{Tr}(-PP^{-1}A) = -\operatorname{Tr}(A) \implies \operatorname{Tr}(A) = 0$.

$\det(A) = \lambda_1\lambda_2\lambda_3 = 0$ implies one of the λ_i is 0. Assume WLOG that $\lambda_1 = 0$.

Since $\operatorname{Tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 = \lambda_2 + \lambda_3 = 0 \implies \lambda_2 = -\lambda_3$. But since they determine the roots of charpoly with real coefficients $\lambda_2 = \bar{\lambda_3}$ also. This means $\operatorname{Re}(\lambda_3) = \operatorname{Re}(\lambda_2) = 0$. Thus the 2 remaining eigenvalues are complex with non-real part. This means both eigenvectors are also complex.

Ex 8.1.19: Similar Matrices

```
sage: K = GF(2)
sage: A = matrix(K, [[1, 0, 1], [0, 1, 0], [1, 0, 1]])
sage: A.charpoly().factor()
(x + 1) * x^2
sage: A.determinant()
0
sage: A = matrix(K, [[1,0,0,0,1],[0,1,0,1,0],[0,0,1,0,0],[0,1,0,1,0],[1,0,0,0,1]])
sage: A.determinant()
0
sage: A.trace()
1
sage: A.charpoly().factor()
(x + 1) * x^4
```

Any diagonal matrix similar to X_1 has the same charpoly. Since it is diagonal, then diagonal$(D - xI) = \{x+1, x, x\}$ (in any order) $\implies$ diagonal$(D) = \{1, 0, 0\}$. Since the trace and determinant are unchanged for similar matrices, we see that $\det(D) \neq \det(A)$ which is a contradiction.

Ex 8.1.20: Complex Eigenvalues of Real Matrix are Conjugate Pairs

$$p(x) \in \mathbb{R}[x] : p(z) = 0 = \overline{p(z)} = p(\overline{z}) \text{ since } p(x) = \overline{p(x)}$$

Ex 8.1.22: Real Eigenvalues of Real Orthogonal Matrix are ± 1

Orthogonal matrix means $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$ or equivalently $Q^T Q = I$.

$$Q\mathbf{v} = \lambda \mathbf{v} \quad \mathbf{v}^T Q^T Q \mathbf{v} = \lambda^2 \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{v} \rangle \implies \lambda = \pm 1$$

$A = \mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(T)$ has Eigenpair $(\mu, \mathbf{x}) \implies T$ has Eigenpair $(\mu, \mathcal{B}\mathbf{x})$

$$\mathbf{v} = \sum x_i \mathbf{v}_i = (\mathbf{v}_1 \cdots \mathbf{v}_n)\mathbf{x}$$
$$(T(\mathbf{v}_1) \cdots T(\mathbf{v}_n)) = (\mathbf{v}_1 \cdots \mathbf{v}_n)A$$

Since $A\mathbf{x} = \mu \mathbf{x}$

$$\begin{aligned} T(\mathbf{v}) &= (T(\mathbf{v}_1) \cdots T(\mathbf{v}_n))\mathbf{x} \\ &= \mu(\mathbf{v}_1 \cdots \mathbf{v}_n)\mathbf{x} \\ &= \mu \mathbf{v} \end{aligned}$$

$A \in \mathbb{F}^n$ has an Eigenbasis $\mathbf{w}_1, ..., \mathbf{w}_n \iff A$ is Diagonalizable

$$AP = (A\mathbf{w}_1 \cdots A\mathbf{w}_n) = (\mathbf{w}_1 \cdots \mathbf{w}_n)\,\mathrm{diag}(\lambda_1, ..., \lambda_n)$$

T is Semisimple with Eigenbasis $(\mathbf{v}_1 \cdots \mathbf{v}_n)P \iff A$ is Diagonalizable

Let $\mathcal{B} = \{\mathbf{v}_1, ..., \mathbf{v}_n\}$ be a basis and $A = \mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(T) = PDP^{-1}$. Since $T(\mathbf{v}_1 \cdots \mathbf{v}_n) = (\mathbf{v}_1 \cdots \mathbf{v}_n)A$ then

$$(T(\mathbf{v}_1) \cdots T(\mathbf{v}_n))P = (\mathbf{v}_1 \cdots \mathbf{v}_n)PD$$

By 7.4.4, we have $TW = TVe = (TV)P$

$$(T(\mathbf{w}_1) \cdots T(\mathbf{w}_n)) = (T(\mathbf{v}_1) \cdots T(\mathbf{v}_n))\mathcal{M}_{\mathcal{B}'}^{\mathcal{B}} = (T(\mathbf{v}_1) \cdots T(\mathbf{v}_n))P$$

but A is defined by $T(\mathbf{v}_1 \cdots \mathbf{v}_n) = (\mathbf{v}_1 \cdots \mathbf{v}_n)A$ so

$$T(\mathbf{w}_1 \cdots \mathbf{w}_n) = (\mathbf{v}_1 \cdots \mathbf{v}_n)AP = (\mathbf{w}_1 \cdots \mathbf{w}_n)D$$

Diagonalization via Eigenspace Decomposition

Distinct Eigenvalues Produce Linearly Independent Eigenvectors

Let $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 = \mathbf{0}$. We want to show $\alpha_1 = \alpha_2 = 0$. Since $A\mathbf{v}_1 = \lambda_1$ and $A\mathbf{v}_2 = \lambda_2$, so $A(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2) = \alpha_1\lambda_1\mathbf{v}_1 + \alpha_2\lambda_2\mathbf{v}_2 = \mathbf{0}$. Subtracting λ_1 times the original equation, we get

$$\alpha_2(\lambda_2 - \lambda_1)\mathbf{v}_2 = \mathbf{0} \Longrightarrow \alpha_2 = 0$$

since $\lambda_1 \neq \lambda_2$. Then we conclude $\alpha_1 = 0$.

Union of Subset of Independent Eigenvectors is Independent

Proposition 8.10: *Suppose $\lambda_1, ..., \lambda_m \in \mathbb{F}$ are distinct eigenvalues of $A \in \mathbb{F}^{n\times n}$, and choose a set S_i of linearly independent eigenvectors in the eigenspace*

$$E_{\lambda_i} = \{\mathbf{v} | (A - \lambda_i I)\mathbf{v} = \mathbf{0}\} = \mathcal{N}(A - \lambda_i I)$$

for all $1 \leq i \leq m$. Then the union of these linearly independent sets $S = S_1 \cup \cdots \cup S_m$ is also linearly independent.

Suppose $A_1 = \{\begin{pmatrix}1 & 0\end{pmatrix}^T\}, A_2 = \{\begin{pmatrix}0 & 1\end{pmatrix}^T\}, A_3 = \{\begin{pmatrix}1 & 1\end{pmatrix}^T\}$. Then $A_i \cap A_j = \{\mathbf{0}\}$ for all $i \neq j$, but $A_1 \cup A_2 \cup A_3$ is linearly dependent. So the result is not generally automatic and needs to be proven for this specific case.

Denote the elements of S as

$$S = \{\mathbf{u}_1^{(1)}, ..., \mathbf{u}_{t_1}^{(1)}, \mathbf{u}_1^{(2)}, ..., \mathbf{u}_{t_2}^{(2)}, ..., \mathbf{u}_1^{(m)}, ..., \mathbf{u}_{t_m}^{(m)}\}$$

We want to show this set is linearly independent, so assume

$$a_1^{(1)}\mathbf{u}_1^{(1)} + \cdots + a_{t_1}^{(1)}\mathbf{u}_{t_1}^{(1)} + a_1^{(2)}\mathbf{u}_1^{(2)} + \cdots + a_{t_2}^{(2)}\mathbf{u}_{t_2}^{(2)} + \cdots + a_1^{(m)}\mathbf{u}_1^{(m)} + \cdots + a_{t_m}^{(m)}\mathbf{u}_{t_m}^{(m)} = \mathbf{0}$$

$$\begin{aligned} M_1 &= \{\mathbf{u}_1^{(1)}, ..., \mathbf{u}_{t_1}^{(1)}\} \\ M_2 &= \{\mathbf{u}_1^{(2)}, ..., \mathbf{u}_{t_2}^{(2)}\} \\ &... \\ M_m &= \{\mathbf{u}_1^{(m)}, ..., \mathbf{u}_{t_m}^{(m)}\} \end{aligned}$$

$$\begin{aligned} \mathbf{v}_1 &= a_1^{(1)}\mathbf{u}_1^{(1)} + \cdots + a_{t_1}^{(1)}\mathbf{u}_{t_1}^{(1)} \in E_{\lambda_1} \\ \mathbf{v}_2 &= a_1^{(2)}\mathbf{u}_1^{(2)} + \cdots + a_{t_2}^{(2)}\mathbf{u}_{t_2}^{(2)} \in E_{\lambda_2} \\ &... \\ \mathbf{v}_m &= a_1^{(m)}\mathbf{u}_1^{(m)} + \cdots + a_{t_m}^{(m)}\mathbf{u}_{t_m}^{(m)} \in E_{\lambda_3} \end{aligned}$$

The elements $\{\mathbf{u}_1^{(i)}, ..., \mathbf{u}_{t_i}^{(i)}\} \in E_{\lambda_i}$ were chosen to be linearly independent from the set E_{λ_i} and we have

$$\begin{aligned} \mathbf{0} &= a_1^{(1)}\mathbf{u}_1^{(1)} + \cdots + a_{t_1}^{(1)}\mathbf{u}_{t_1}^{(1)} + a_1^{(2)}\mathbf{u}_1^{(2)} + \cdots + a_{t_2}^{(2)}\mathbf{u}_{t_2}^{(2)} + \cdots + a_1^{(m)}\mathbf{u}_1^{(m)} + \cdots + a_{t_m}^{(m)}\mathbf{u}_{t_m}^{(m)} \\ &= (a_1^{(1)}\mathbf{u}_1^{(1)} + \cdots + a_{t_1}^{(1)}\mathbf{u}_{t_1}^{(1)}) + (a_1^{(2)}\mathbf{u}_1^{(2)} + \cdots + a_{t_2}^{(2)}\mathbf{u}_{t_2}^{(2)}) + \cdots + (a_1^{(m)}\mathbf{u}_1^{(m)} + \cdots + a_{t_m}^{(m)}\mathbf{u}_{t_m}^{(m)}) \\ &= \mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m \\ &= \sum_{i=1}^{m} \mathbf{v}_i \end{aligned}$$

Now we show all $\mathbf{v}_i = \mathbf{0}$.

WLOG suppose $\mathbf{v}_1 \neq 0$, then $\mathbf{v}_1 = -(\mathbf{v}_2 + \cdots + \mathbf{v}_m) \neq 0$.

Now select from the set $\{\mathbf{v}_2, ..., \mathbf{v}_m\}$, a subset that form a basis of $W = \text{span}\{\mathbf{v}_2, ..., \mathbf{v}_m\}$. Relabelling as necessary, denote these by $\{\mathbf{v}_2, ..., \mathbf{v}_l\}$ where $l = \dim(W)$. Each of these $\mathbf{v}_i \in E_{\lambda_i}$ is an eigenvector with eigenvalue λ_i.

Since the eigenvalues are distinct $\lambda_1 \neq \lambda_i$ for all $2 \leq i \leq l$. By the previous result that "distinct eigenvalues produce linearly independent eigenvectors", this means that $b_2 = \cdots = b_l = 0 \implies \mathbf{v}_1 = \mathbf{0}$.

Reapplying the same logic, we prove all $\mathbf{v}_i = \mathbf{0}$. This means that

$$\mathbf{v}_i = a_1^{(i)}\mathbf{u}_1^{(i)} + \cdots + a_{t_i}^{(i)}\mathbf{u}_{t_i}^{(i)} = \mathbf{0}$$

Since each $\mathbf{u}_j^{(i)}$ is linearly independent within $\{\mathbf{u}_1^{(i)}, ..., \mathbf{u}_{t_i}^{(i)}\} \in E_{\lambda_i}$, this means $a_1^{(i)} = \cdots = a_{t_i}^{(i)} = 0$ for all i.

Therefore S is linearly independent.

Diagonalizable Condition $V = E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_m}(A)$

Let $\lambda_1, ..., \lambda_m$ denote the eigenvalues of A. A is diagonalizable iff

$$\sum_{i=1}^{m} \dim E_{\lambda_i}(A) = n$$

If $\mathcal{B}_i$ is a basis of $E_{\lambda_i}(A)$ then $\mathcal{B} = \bigcup \mathcal{B}_i$ and

$$V = E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_m}(A)$$

To prove this, use the previous proposition and the dimension theorem.

T is Semisimple $\implies (T - \lambda_1 I)\cdots(T - \lambda_m I) = O$

Since T is semisimple, we have $V = E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_m}(A)$. So we just have to show that for $\mathbf{v}_i \in E_{\lambda_i}(A)$ that

$$(T - \lambda_1 I)\cdots(T - \lambda_m I)\mathbf{v}_i = \mathbf{0}$$

{#eq:tl0}

Note that $(A - xI)(A - yI) = A - (x + y)I + xyI = (A - yI)(A - xI)$. Therefore[eq:tl0?] can be factored in the form $C(T - \lambda_i I)$

$$\implies C(T - \lambda_i I)\mathbf{v}_i = \mathbf{0}$$

Alternative Proof of T is Semisimple $\implies (T - \lambda_1 I)\cdots(T - \lambda_m I) = O$

$$\begin{aligned} f(A) &= a_n A^n + \cdots + a_1 A + a_0 I \\ &= a_n (PDP^{-1})^n + \cdots + a_1 (PDP^{-1}) + a_0 (PP^{-1}) \\ &= a_n PD^n P^{-1} + \cdots + a_1 PDP^{-1} + a_0 PP^{-1} \\ &= P(a_n D^n + \cdots + a_1 D + a_0 I)P^{-1} \\ &= Pf(D)P^{-1} \end{aligned}$$

Now let $g(X) = (X - \lambda_1 I)\cdots(X - \lambda_m I)$ then

$$g(A) = Pg(D)P^{-1}$$

but $D = \operatorname{diag}(\lambda_1, ..., \lambda_m) \implies (D - \lambda_1)\cdots(D - \lambda_m) = O$ so

$$g(A) = O$$

T is Semisimple $\impliedby (T - \lambda_1 I)\cdots(T - \lambda_m I) = O$

$P \circ Q = O$ **and** $\ker(P) \cap \ker(Q) = \{\mathbf{0}\} \implies V = \ker(P) \oplus \ker(Q)$

2. $PQ = O$
3. $\ker(P) \cap \ker(Q) = \{\mathbf{0}\}$

$$\dim(V) = \dim\ker(Q) + \dim\operatorname{im}(Q)$$

but $P \circ Q = O$ so $\operatorname{im}(Q) \subseteq \ker(P)$ and so

$$\dim\operatorname{im}(Q) = \dim(V) - \dim\ker(Q) \leq \dim\ker(P)$$

$$\implies \dim(V) \leq \dim\ker(Q) + \dim\ker(P)$$

Since $\ker(P) \cap \ker(Q) = \{\mathbf{0}\}$, we have $\dim(\ker(P) \cap \ker(Q)) = 0$. By the Grasmann intersection formula

$$\begin{aligned}\dim(\ker(P) + \ker(Q)) &= \dim\ker(P) + \dim\ker(Q) - \dim(\ker(P) \cap \ker(Q)) \\ &= \dim\ker(P) + \dim\ker(Q)\end{aligned}$$

So we see that $\dim(V) \leq \dim(\ker(P) + \ker(Q))$, but $\ker(P) + \ker(Q) \subseteq V$ so

$$V = \ker(P) + \ker(Q)$$

Lastly since $\ker(P) \cap \ker(Q) = \{\mathbf{0}\}$, we have

$$V = \ker(P) \oplus \ker(Q)$$

Generalizing Above to $V = \ker(T_1) \oplus \cdots \oplus \ker(T_n)$

Lemma. *Suppose $T_1, ..., T_n : V \to V$ are linear mappings that satisfy the following properties:*

1. $T_1 \circ \cdots \circ T_n = O$
2. $T_i \circ T_j = T_j \circ T_i$ for all i, j
3. $\ker(T_i) \cap \ker(T_{i+1} \circ \cdots \circ T_n) = \{\mathbf{0}\}$ for all i

Then $V = \ker(T_1) \oplus \cdots \oplus \ker(T_n)$.

Proof. Let $P = T_1$ and $Q = T_2 \circ \cdots \circ T_n$, then

$$V = \ker(T_1) \oplus \ker(T_2 \circ \cdots \circ T_n)$$

Now set $V' = \ker(T_2 \circ \cdots \circ T_n), P = T_2, Q = T_3 \circ \cdots \circ T_n$ and we get

$$V = \ker(T_1) \oplus \ker(T_2) \oplus \ker(T_3 \circ \cdots \circ T_n)$$

Iterating we arrive at the conclusion

$$V = \ker(T_1) \oplus \cdots \oplus \ker(T_n)$$

T is Semisimple

Condition (1) is our starting assumption $(T - \lambda_1 I) \cdots (T - \lambda_m I) = O$.

For (2), we can also see that $(T - \lambda_1 I)(T - \lambda_2 I) = (T - \lambda_2 I)(T - \lambda_1 I)$.

Finally to prove (3), let $\mathbf{x} \in \ker(T - \lambda_i I)$ with $\mathbf{x} \neq 0$, then $(T - \lambda_j I)\mathbf{v} = (\lambda_i - \lambda_j)\mathbf{v}$ and so

$$(T - \lambda_{i+1} I) \cdots (T - \lambda_m I)\mathbf{x} = (\lambda_i - \lambda_{i+1}) \cdots (\lambda_i - \lambda_m)\mathbf{x} \neq 0$$

since λ_i are distinct from all λ_j.

Then by the previous lemma

$$V = E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_m}(A)$$

Exercises 8.3

Ex 8.3.6

Computing the eigenvalues gives 2 distinct real values.

Ex 8.3.7

Let λ_1, λ_2 be the eigenvalues for A. Then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \lambda_1 \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \lambda_2 \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$ax_1 + by_1 = \lambda_1 x_1$$

$$ax_2 + by_2 = \lambda_2 x_2$$

(x, y) is in the first quadrant means $\frac{x}{y} > 0$. Likewise (x, y) is in the second/fourth quadrant means $\frac{x}{y} < 0$. Rewriting both equations, we get

$$\frac{x_1}{y_1} = -\frac{b}{a - \lambda_1}$$
$$\frac{x_2}{y_2} = -\frac{b}{a - \lambda_2}$$

Noting that $\text{Tr}(A) = a + d = \lambda_1 + \lambda_2$ and $\det(A) = ad - bc = \lambda_1\lambda_2$ we see

$$\frac{x_1 x_2}{y_1 y_2} = -\frac{b}{c} < 0$$

So one fraction is in the first quadrant, and another in the second.

Ex 8.3.10

$$\mathbb{F}^{n\times n} = \mathbb{F}_s^{n\times n} \oplus \mathbb{F}_{ss}^{n\times n}$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

When $A \in \mathbb{F}_s^{n\times n}$ then $\mathbb{T}(A) = A$ so $\lambda_1 = 1$.

When $A \in \mathbb{F}_{ss}^{n\times n}$ then $\mathbb{T}(A) = -A$ so $\lambda_1 = -1$.

Now the basis for $\mathbb{F}_s^{n\times n}$ and $\mathbb{F}_{ss}^{n\times n}$ are basis for $E_{\lambda_1}(\mathbb{T})$ and $E_{\lambda_s}(\mathbb{T})$, so $\mathbb{T}$ is semisimple.

Ex 8.3.15

charpoly$(U) = (x - \lambda_1)\cdots(x - \lambda_n)$ where $\lambda_1, ..., \lambda_n$ are the diagonals of U. Each space $\dim E_{\lambda_i}(U) \geq 1 \implies \dim E_{\lambda_1} = \cdots = \dim E_{\lambda_n} = 1$ and so $\sum \dim E_{\lambda_i} = n \implies V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_n}$ and thus U is diagonalizable.

Ex 8.3.17/18

$A = PD_1P^{-1}, B = PD_2P^{-1} \implies AB = PD_1D_2P^{-1} = PD_2D_1P^{-1} = BA$.

Let $\mathbf{v} \in E_\lambda(A)$, then $BA\mathbf{v} = \lambda B\mathbf{v} = AB\mathbf{v} \implies B\mathbf{v} \in E_\lambda(A)$.

Our goal is to find a common eigenbasis between A and B. Since A and B are diagonalizable, they both admit eigenbasis such that

$$\begin{aligned} V &= E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_l}(A) \\ &= E_{\mu_1}(B) \oplus \cdots \oplus E_{\mu_m}(B) \end{aligned}$$

viewing $A, B : V \to V$ as linear maps, we can take the restriction of

$$B|_{E_{\lambda_i}(A)} : E_{\lambda_i}(A) \to E_{\lambda_i}(A)$$

which is a valid restriction since $BE_{\lambda_i}(A) \subseteq E_{\lambda_i}(A)$. Now let

$$\begin{aligned} V(\lambda, \mu) &= \{\mathbf{v} \in E_{\lambda_i}(A) | (B - \mu)\mathbf{v} = \mathbf{0}\} \\ &= \{\mathbf{v} \in V | (A - \lambda)\mathbf{v} = (B - \mu)\mathbf{v} = \mathbf{0}\} \end{aligned}$$

$$\implies E_{\lambda_i}(A) = V(\lambda_i, \mu_1) \oplus \cdots \oplus V(\lambda_i, \mu_m)$$
$$V_{\lambda,\mu\in\mathbb{F}} = \bigoplus V(\lambda, \mu)$$

and since $\dim V = n$, we can pick an independent eigenbasis for both A, B.

Ex 8.3.19

U is diagonalizable when the $\sum \dim E_{\lambda_i}(U) = n$. Assume WLOG that λ is repeated twice on the diagonal for rows i, j, then we want that $\dim E_\lambda(U) = 2$. This means the row reduced form of $(U - \lambda I)$, must have zero rows for i, j which is the same as saying $\dim(U - \lambda I) = \dim E_\lambda(U) = 2$.

Based off how row reduction works, see see that all vertical entries above λ should be 0, otherwise $(U - \lambda I)$ has another independent row vector.

Flag Variety: Triangulation of Operators and Matrices

- Not every matrix is diagonalizable
- What we prove now is that for any operator on the complex finite dimensional vector space, there exists a basis in which its matrix is upper triangular.

Let V be a finite dimensional vector space.

Flag

Let V be a finite dimensional vector space. A flag in V is a sequence of nested subspaces

$$V_1 \subset V_2 \subset \cdots \subset V_n = V$$

such that $\dim V_k = k$.

A basis $\mathcal{B}$ of $V, \mathcal{B} = \{\mathbf{v}_1, ..., \mathbf{v}_n\}$ is called compatible with a given flag if $V_1 = \text{span}\{\mathbf{v}_1\}, ..., V_n = \{\mathbf{v}_1, ..., \mathbf{v}_n\}$.

T-invariant flag

Let $T : V \to V$ be a linear operator. We say that a flag is T-invariant if $\forall k = 1, ..., n$, the subspace V_k is invariant under T

$$T(V_k) \subset V_k$$

Let $\mathcal{B} = \{\mathbf{v}_1, ..., \mathbf{v}_n\}$ be a basis of V compatible with a T-invariant flag. What can we say about $T_{\mathcal{B}}$?

$T(\mathbf{v}_1) \in V_1 \implies T(\mathbf{v}_1) = a_{11}\mathbf{v}_1$ by the defn of T-invariance. The first basis vector is therefore an eigenvector for our linear operator T.

Continuing we get

$$T(\mathbf{v}_2) \in V_2 \implies T(\mathbf{v}_2) = a_{12}\mathbf{v}_1 + a_{22}\mathbf{v}_2$$
$$T(\mathbf{v}_3) \in V_3 \implies T(\mathbf{v}_3) = a_{13}\mathbf{v}_1 + a_{23}\mathbf{v}_2 + a_{33}\mathbf{v}_3$$

so putting this in a matrix we get

$$T_{\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ 0 & a_{22} & a_{23} & \cdots \\ 0 & 0 & a_{33} & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}$$

Conclusion: $T_{\mathcal{B}}$ is upper triangular.

Theorem: Every linear operator $T : V \to V$ on a complex finite-dimensional vector space V has an invariant flag

Corollary: for a square complex matrix A there exist an upper-triangular matrix S and an invertible matrix C such that

$$A = CSC^{-1}$$

Proof of corollary: Let $\mathcal{B}$ be a basis compatible with an A-invariant flag, then we get $A = C_{\mathcal{B}}^{\mathcal{E}} T_{\mathcal{B}} C_{\mathcal{E}}^{\mathcal{B}}$.

Proof by induction on $n = \dim V$: $n = 1$ then $V_1 = V$, then we see $T(V)$ is automatically invariant.

Now assume theorem holds for operators on $(n-1)$-dimensional vector spaces over $\mathbb{C}$. We know that any operator on a complex vector space of $\dim n \geq 1$ has an eigenvector: $T(\mathbf{v}) = \lambda\mathbf{v}$.

Set $\mathbf{v}_1 = \mathbf{v}$, take $V_1 = \text{span}\{\mathbf{v}_1\}$, then since it is an eigenspace, it is invariant and $T(\mathbf{v}_1) \subset V_1$.

By the 3rd homomorphism theorem, there exists an operator T'

$$T' : V/V_1 \to V/V_1$$

such that the following diagram commutes

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \downarrow P & & \downarrow P \\ V/V_1 & \xrightarrow{T'} & V/V_1 \end{array}$$

Now let $W = V/V_1$, then $\dim W = n-1$. By induction then T' has an invariant flag in W. Let $\{\mathbf{w}_1, ..., \mathbf{w}_{n-1}\}$ be a basis of W, compatible with T'-invariant flag. Then we can write these in terms of equivalence classes $\mathbf{w}_1 = [\mathbf{v}_2], ..., \mathbf{w}_{n-1} = [\mathbf{v}_n]$. Now we have a basis in the 1-dim space V_1, and a basis in the quotient space W, and we have the result that describes the relation basis in V with the basis of the subspace and the basis of the quotient space.

Then $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ is a basis of V, because $\mathbf{v}_1$ is a basis in the subspace, and the equivalence classes of $\mathbf{v}_2, ..., \mathbf{v}_n$ form a basis of the quotient space W.

Claim: $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ is a basis compatible with a T-invariant flag

$$V_1 = \text{span}\{\mathbf{v}_1\}, ..., V_n = \text{span}\{\mathbf{v}_1, ..., \mathbf{v}_n\}$$

We need to prove $T(\mathbf{v}_k)$ is a linear combination of $\mathbf{v}_1, ..., \mathbf{v}_k$ for $k = 1, ..., n$.

We know this holds for $k = 1$ since $T(\mathbf{v}_1) = \lambda \mathbf{v}_1$.

We know $T'([\mathbf{v}_k])$ is a linear combination of $[\mathbf{v}_2], ..., [\mathbf{v}_k]$ since $\{\mathbf{w}_1, ..., \mathbf{w}_{n-1}\}$ form a basis of W.

$$T'([\mathbf{v}_k]) = c_2[\mathbf{v}_2] + \cdots + c_k[\mathbf{v}_k]$$

$$T'([\mathbf{v}_k]) = [T(\mathbf{v}_k)] \implies [T(\mathbf{v}_k)] - c_2[\mathbf{v}_2] - \cdots - c_k[\mathbf{v}_k] = [T(\mathbf{v}_k) - c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k] = [0]$$

$$T(\mathbf{v}_k) - c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k \in V_1 \implies \exists c_1 \in \mathbb{C} : T(\mathbf{v}_k) - c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k = c_1\mathbf{v}_1$$

so we see $T(\mathbf{v}_k) = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$.

Proof of Cayley-Hamilton

$n = 1$ is trivial, so assume result is true for $n-1$ given $n > 1$.

$(\lambda_1, \mathbf{v}_1)$ is an eigenpair for A. Extend $\mathbf{v}_1$ to a basis $\mathcal{B}$ of V. Then A is similar to $B = \mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(T_A)$ and $p_A(x) = p_B(x)$.

$$B = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & & & \\ \vdots & & B_1 & \\ 0 & & & \end{pmatrix}$$

Now we note the eigenvalues of B_1 are in $\mathbb{F}$ since

$$\begin{aligned} p_A(x) &= p_B(x) \\ &= \det(B - xI_n) \\ &= (\lambda_1 - x)\det(B_1 - xI_{n-1}) \\ &= (\lambda_1 - x)p_{B_1}(x) \end{aligned}$$

and we see they are the eigenvalues $\{\lambda_2, ..., \lambda_n\}$ of B.

We equivalently calculate whether $p_B(B) = O \implies p_A(A) = O$. We first note a general fact about matrices

$$\begin{pmatrix} c_1 & * & \cdots & * \\ 0 & & & \\ \vdots & & C_1 & \\ 0 & & & \end{pmatrix} \begin{pmatrix} c_2 & * & \cdots & * \\ 0 & & & \\ \vdots & & C_2 & \\ 0 & & & \end{pmatrix} = \begin{pmatrix} c_1c_2 & * & \cdots & * \\ 0 & & & \\ \vdots & & C_1C_2 & \\ 0 & & & \end{pmatrix}$$

We now wish to calculate $p_B(B)$ by lifting $p_B(x)$ with the map

$$\phi : \mathbb{F}[x] \to \mathbb{F}[B]$$

$$\begin{aligned}
\phi(p_B(x)) &= \phi((-1)^n(x-\lambda_1)\cdots(x-\lambda_n)) \\
&= (-1)^n(B-\lambda_1 I_n)p_{B_1}(B_1) \\
&= (-1)^n \begin{pmatrix} 0 & * & \cdots & * \\ 0 & & & \\ \vdots & & B_1-\lambda_1 I_{n-1} & \\ 0 & & & \end{pmatrix} \begin{pmatrix} * & * & \cdots & * \\ 0 & & & \\ \vdots & & p_{B_1}(B_1) & \\ 0 & & & \end{pmatrix} \\
&= (-1)^n \begin{pmatrix} 0 & * & \cdots & * \\ 0 & & & \\ \vdots & & B_1-\lambda_1 I_{n-1} & \\ 0 & & & \end{pmatrix} \begin{pmatrix} * & * & \cdots & * \\ 0 & & & \\ \vdots & & O & \\ 0 & & & \end{pmatrix} \\
&= O
\end{aligned}$$

since $p_{B_1}(B_1) = O$ by the induction hypothesis.

Hermitian Principal Axis Theorem (also called Spectral Theorem)

$V = \ker(T) \oplus \operatorname{im}(T)$

We prove $\operatorname{im}(T) \subset \ker(T)^\perp$. Let $\mathbf{w} \in \ker(T)$ and $T(\mathbf{x}) \in \operatorname{im}(T)$ then

$$\langle \mathbf{w}, T(\mathbf{x}) \rangle = \langle T(\mathbf{w}), \mathbf{x} \rangle = \mathbf{0}$$

so $T(x) \in \ker(T)^\perp$ which means $\operatorname{im}(T) \subset \ker(T)^\perp$.

Lets note two theorems. The first concerns a subspace and its orthogonal complement, $V = W \oplus W^\perp$. Secondly the rank-nullity theorem $\dim V = \dim\ker(T) + \dim\operatorname{im}(T)$.

Now apply both theorems to get

$$\begin{aligned}
\dim V &= \dim\ker(T) + \dim\ker(T)^\perp \\
&= \dim\ker(T) + \dim\operatorname{im}(T) \\
\implies \dim\ker(T)^\perp &= \dim\operatorname{im}(T)
\end{aligned}$$

and since $\operatorname{im}(T) \subset \ker(T)^\perp$ so

$$\ker(T)^\perp = \operatorname{im}(T)$$

Principal Axis Theorem: Orthonormal Basis of V from Eigenvectors of T

Assume true for $n-1$, let $\dim V = n$. Let $\lambda \in \mathbb{R}$ and $S = T - \lambda I$.

If $V = \ker(S)$ then we are done, so assume $V \neq \ker(S)$.

S is self-adjoint since

$$\begin{aligned}
\langle S(\mathbf{v}_1), \mathbf{v}_2 \rangle &= \langle T\mathbf{v}_1 - \lambda\mathbf{v}_1, \mathbf{v}_2 \rangle \\
&= \langle T\mathbf{v}_1, \mathbf{v}_2 \rangle - \lambda\langle \mathbf{v}_1, \mathbf{v}_2 \rangle \\
&= \langle \mathbf{v}_1, T\mathbf{v}_2 \rangle - \langle \mathbf{v}_1, \lambda\mathbf{v}_2 \rangle \\
&= \langle \mathbf{v}_1, S(\mathbf{v}_2) \rangle
\end{aligned}$$

so from the previous proposition we see

$$V = \ker(S) \oplus \operatorname{im}(S)$$

Observe that $T(\ker(S)) \subset \ker(S)$ and $T(\operatorname{im}(S)) \subset \operatorname{im}(S)$ so we can construct restrictions of T to both of these subspaces.

Since $0 < \dim\ker(S) \leq n-1$ and $0 < \dim\operatorname{im}(S) \leq n-1$ we can apply the induction to both of these subspaces.

Formula for P_λ when A is Symmetric (Alternative)

This is similar to the lagrange interpolation trick.

$$P_\lambda = \prod_{\mu \neq \lambda} \frac{A - \mu I}{\lambda - \mu}$$

Let the eigenpairs be $(\lambda, \mathbf{v}_\lambda)$ and $(\mu, \mathbf{v}_\mu)$. Applying P_λ we see

$$\begin{aligned} P_\lambda(\mathbf{v}_\lambda) &= \prod_{\mu \neq \lambda} \frac{A\mathbf{v}_\lambda - \mu \mathbf{v}_\lambda}{\lambda - \mu} \\ &= \prod_{\mu \neq \lambda} \frac{(\lambda - \mu)\mathbf{v}_\lambda}{\lambda - \mu} \\ &= \mathbf{v}_\lambda \end{aligned}$$

$$\begin{aligned} P_\lambda(\mathbf{v}_\mu) &= \prod_{\mu \neq \lambda} \frac{A\mathbf{v}_\mu - \mu \mathbf{v}_\mu}{\lambda - \mu} \\ &= \prod_{\mu \neq \lambda} \frac{(\mu - \mu)\mathbf{v}_\lambda}{\lambda - \mu} \\ &= \mathbf{0} \end{aligned}$$

Therefore we get that $P_\lambda^2 = P_\lambda$ by the above identities.

Then also for any two distinct eigenvalues we have $P_\lambda P_\mu = P_\mu P_\lambda = O$.

Lastly since $A = A^T$ is symmetric, it follows from our definition that $P_\lambda = P_\lambda^T$.

The formula given in the book when we have the orthonomal eigenvector $\mathbf{u}_\lambda$ is simply $P_\lambda = \mathbf{u}_\lambda \mathbf{u}_\lambda^T$ which is obtained by examining the formula $P_\lambda(\mathbf{v}) = \langle \mathbf{u}_\lambda, \mathbf{v} \rangle \mathbf{u}_\lambda = \mathbf{u}_\lambda(\mathbf{u}_\lambda^T \mathbf{v}) = (\mathbf{u}_\lambda \mathbf{u}_\lambda^T)\mathbf{v}$.

Exercises 8.4

Ex 8.4.3

False proof of Cayley-Hamilton: $p_A(A) = \det(A - A) = 0$

Ex 8.4.4

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

A is nilpotent means $A^2 = O$, so $A^2 - \mathrm{Tr}(A)A + \det(A)I = O \implies \det(A)I = \mathrm{Tr}(A)A$.

```
sage: var("a b c d")
(a, b, c, d)
sage: A = matrix([[a, b], [c, d]])
sage: C = (-A.trace()*A + A.det()*matrix.identity(2))
sage: e1 = C[0][0].expand() == 0
sage: e2 = C[0][1].expand() == 0
sage: e3 = C[1][0].expand() == 0
sage: e4 = C[1][1].expand() == 0
sage: e1
-a^2 - b*c == 0
sage: e2
-a*b - b*d == 0
sage: e3
-a*c - c*d == 0
sage: e4
-b*c - d^2 == 0
sage: e1 - e4
```

```
-a^2 + d^2 == 0
sage: # so a = +/-d
sage: e2 + b*d
-a*b == b*d
sage: # => a = -d
sage: e1.subs({-a^2: -a*(-d)})
-b*c + a*d == 0
sage: # so det(A) == 0
sage: # and since a = -d => a + d = Tr(A) == 0
```

Ex 8.4.5

$A^k = O$ and let $(\lambda, \mathbf{v})$ be an eigenpair for A, then $A^k\mathbf{v} = \lambda^k\mathbf{v} = 0 \implies \lambda = 0$.

And if all eigenvalues are 0, then $p(x) = x^n \implies p(A) = A^n = O$.

Ex 8.4.6

Assume $A = PDP^{-1}$ is diagonalizable, then $A^k = PD^kP^{-1} = O \implies D = O$ and so $A = O$.

Ex 8.4.7

$A^k = O \iff A^n\mathbb{F}^n = \{\mathbf{0}\}$ so $\dim \mathcal{N}(A) > 0$. Let $\mathbf{u}_1, ..., \mathbf{u}_s$ be a basis of $\mathcal{N}(A)$ and extend it to a basis of $\text{col}(A)$, then we see

$$\mathbb{F}^n = \mathbf{u}_1 \oplus \cdots \oplus \mathbf{u}_s \oplus \mathbf{w}_1 \oplus \cdots \oplus \mathbf{w}_t$$

now let k be minimal such that $A^k = O$ and $k > n$, then we have a chain

$$A^k\mathbb{F}^n = \{0\} \subset \cdots \subset A\mathbb{F}^n \subset \mathbb{F}^n$$

with

$$\dim A^k\mathbb{F}^n = 0 < \cdots < \dim A\mathbb{F}^n \dim \mathbb{F}^n = n$$

viewing $T_A(\mathbf{v}) = A\mathbf{v}$, we see $\dim A\mathbb{F}^n = \dim T_A = \text{rank}(A)$.

But there are n maximum steps, so $k < n$.

Ex 8.4.12

A forms a 1d vector space over $\mathbb{F}^{n\times n}$, so if $A^2 \notin \text{span}\{A\}$ then $\{A, A^2\}$ are independent and so on. So $T^0, ..., T^{k-1}$ are independent. Also we can just view powers of T as a basis for a subspace in $\mathcal{L}(V, V)$.

Note that $\mathcal{L}(V, W) = (\dim V)(\dim W)$ although we only use the weaker property that $\mathcal{L}(V, V)$ is finite dimension given that V is finite dimension.

Ex 8.5.2

Proposition 8.22. A linear mapping $T : V \to V$ is self-adjoint iff its matrix with respect to an arbitrary orthonormal basis of V is symmetric.

$$\langle \mathbf{x}, A\mathbf{y}\rangle = \langle A\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, A^T\mathbf{y}\rangle$$

Then using the orthonormal basis $\mathbf{u}_1, ..., \mathbf{u}_n$ we see

$$A\mathbf{u}_j = a_{1j}\mathbf{u}_1 + \cdots + a_{nj}\mathbf{u}_n$$

$$A^T\mathbf{u}_j = b_{1j}\mathbf{u}_1 + \cdots + b_{nj}\mathbf{u}_n$$

And then evaluating $\langle \mathbf{u}_i, A\mathbf{u}_j\rangle = \langle \mathbf{u}_j, A^T\mathbf{u}_j\rangle$, we see $A = A^T$.

Ex 8.5.6

$$W = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\} \quad \mathbf{u}_3 \in W^\perp \quad P_i = \mathbf{u}_i\mathbf{u}_i^T$$

$$B = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 \text{ where } \lambda_3 = \det(B)/(\lambda_1\lambda_2)$$

Ex 8.5.7

Product of symmetric matrices is only symmetric when $AB = BA$ since $AB = A^TB^T = (BA)^T$.

Rearrange A and B to get $D = Q^TAQ = R^TBR$ so $A = QR^TBRQ^T = (QR^T)B(QR^T)^T$ so is true.

Ex 8.5.8

Let $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ be eigenpairs of A. Since $\det(A) = 0$, one of the eigenvalues of A is 0. Because $\mathbf{v}_1, \mathbf{v}_2$ lie in the image of T_A, so $\lambda_1, \lambda_2 \neq 0$ and $\lambda_3 = 0$. So we see

$$A = \lambda_1 P_1 + \lambda_2 P_2$$

Making $\mathbf{v}_1, \mathbf{v}_2$ into orthonormal vectors and constructing P_1, P_2, we see that

$$\mathrm{Tr}(A) = \lambda_1 + \lambda_2 = 2\lambda_1 + 3\lambda_2 = 4$$

then we get $\lambda_1 = \lambda_2$

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad \lambda_1 = \lambda_2 = 2 \quad \lambda_3 = 0 \quad \mathbf{v}_3 = \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}^T$$

Ex 8.5.11

$$W = \mathrm{span}\{\mathbf{w}_1, ..., \mathbf{w}_s\} \quad W^\perp = \mathrm{span}\{\mathbf{v}_1, ..., \mathbf{v}_t\}$$

with $s+t = n$ and $P_W = \mathbf{w}_1\mathbf{w}_1^T + \cdots + \mathbf{w}_s\mathbf{w}_s^T$ where all $\mathbf{w}_i, \mathbf{v}_i$ are normal. Then $Q = \begin{pmatrix} \mathbf{w}_1 & \cdots & \mathbf{w}_s & \mathbf{v}_1 & \cdots & \mathbf{v}_t \end{pmatrix}, D_1 = \mathrm{diag}(\underbrace{1, ..., 1}_{s}, \underbrace{0, ..., 0}_{t})$ and $D_2 = \mathrm{diag}(\underbrace{0, ..., 0}_{s}, \underbrace{1, ..., 1}_{t})$. Then $P_W = QD_1Q^T, P_{W^\perp} = QD_2Q^T$.

Ex 8.5.12

$\det(A) = \lambda_1\lambda_2\lambda_3, \mathrm{Tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 = \lambda_3 \implies \lambda_1 = -\lambda_2$ so $\lambda_1\lambda_2 < 0$. Since $\lambda_3 \in \mathbb{R}$ so $\lambda_3^2 > 0$ or $\det(A)\,\mathrm{Tr}(A) = \lambda_1\lambda_2\lambda_3^2 < 0$.

Ex 8.5.13

For (i), factor the determinant and notice that $Tr(A)$ is a term, so $\det(A) = 0$.

The eigenvalues for A are $0, \mathrm{Tr}(A)$. The eigenvector for $\lambda_1 = \mathrm{Tr}(A)$ is $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$ and the rest is easy.

Ex 8.5.23

$$\det(R - xI) = (x - \cos(\theta))^2 + \sin^2(\theta) \implies \lambda = \cos(\theta) \pm i\sin(\theta)$$

Ex 8.5.26 (ii)

$$\langle U\mathbf{v}, U\mathbf{v}\rangle = \langle \lambda\mathbf{v}, \lambda\mathbf{v}\rangle = \langle \mathbf{v}, U^HU\mathbf{v}\rangle = \langle \mathbf{v}, \mathbf{v}\rangle \implies |\lambda| = 1$$

so since $\det(U) = 1$, either $\lambda_1 = \lambda_2 = 1$ or $\lambda_1 = \lambda_2 = -1$.

Ex 8.5.32

$$\Phi(a_1\mathbf{u}_1 + \cdots + a_n\mathbf{u}_n) = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix}^T$$

where $\{\mathbf{u}_1, ..., \mathbf{u}_n\}$ is the basis for V.

Eigenvectors within the same eigenspace are easily orthogonalized using the Gram-Schmidt method.

Now let $\mathbf{v}_1 \in E_{\lambda_1}$ and $\mathbf{v}_2 \in E_{\lambda_2}$ where $\lambda_1 \neq \lambda_2$, then

$$\begin{aligned} \lambda_1\langle \mathbf{v}_1, \mathbf{v}_2\rangle &= \langle T(\mathbf{v}_1), \mathbf{v}_2\rangle \\ &= \langle \mathbf{v}_1, T(\mathbf{v}_2)\rangle \\ &= \lambda_2\langle \mathbf{v}_1, \mathbf{v}_2\rangle \end{aligned}$$

$$\implies (\lambda_1 - \lambda_2)\langle \mathbf{v}_1, \mathbf{v}_2\rangle = 0$$

but $\lambda_1 \neq \lambda_2$ so $\langle \mathbf{v}_1, \mathbf{v}_2\rangle = 0$.

Field Extensions

$\dim V/W = \deg(f)$

$W = \langle f(x) \rangle$ and let $h(\alpha) = a_0 + a_1\alpha + \cdots + a_k\alpha^k = \bar{\mathbf{0}}$ where $k = \deg(f) - 1$. This means that $h(x) \in W$. But since $h = gf$ for some $g \in K[x] \implies g = 0$. So $\{1, \alpha, ..., \alpha^k\}$ are independent.

To show that $\{1, \alpha, ..., \alpha^k\}$ span, for any $g \in K[x]$, then by the division algo $g = qf + r$ and $\deg(r) < \deg(f)$. Then $\bar{r} \in \text{span}\{1, ..., \alpha^k\}$.

V/W is a field

Let f be irreducible.

If $\bar{g} \neq \bar{0}$ and since f is irreducible, then f and g share no common factor. There are $a, b \in K[x]$ such that

$$af + bg = 1$$

$$\implies \bar{b}\bar{g} \equiv \bar{1} \pmod{f}$$

since $\overline{bg} = \bar{b}\bar{g}$.